

EXERCISE 1

1. When is the Cout output of the binary counter turned on (logic 1)?

The Cout output of a binary counter is activated (goes to logic 1) when the counter reaches its maximum value. In this case, the counter is 4 bits. Therefore, Cout is activated when the counter reaches the value 15 (1111 in binary), indicating an "overflow" of the counter.

2. Drive the reset control input of the counter to 1. What do you observe?

When the Reset is set to logic 1, the counter resets, and all outputs (Q0, Q1, Q2, Q3) are immediately set to 0, regardless of the state of the Clk or Enable inputs.

3. Drive the enable control input of the counter to 0, while the reset is set to 0. What do you observe?

When Enable is set to logic 0, the counter stops counting, meaning it remains static and does not respond to clock pulses. The outputs (Q0, Q1, Q2, Q3) remain in their current state until Enable is set back to 1.